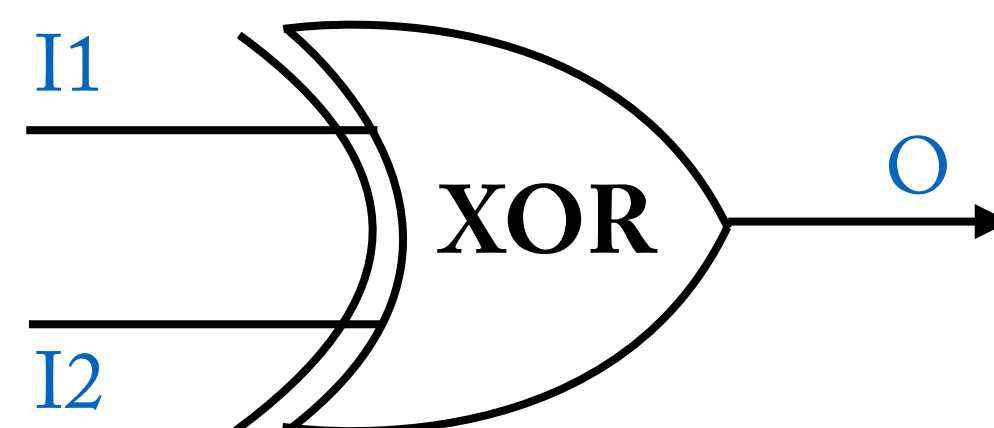
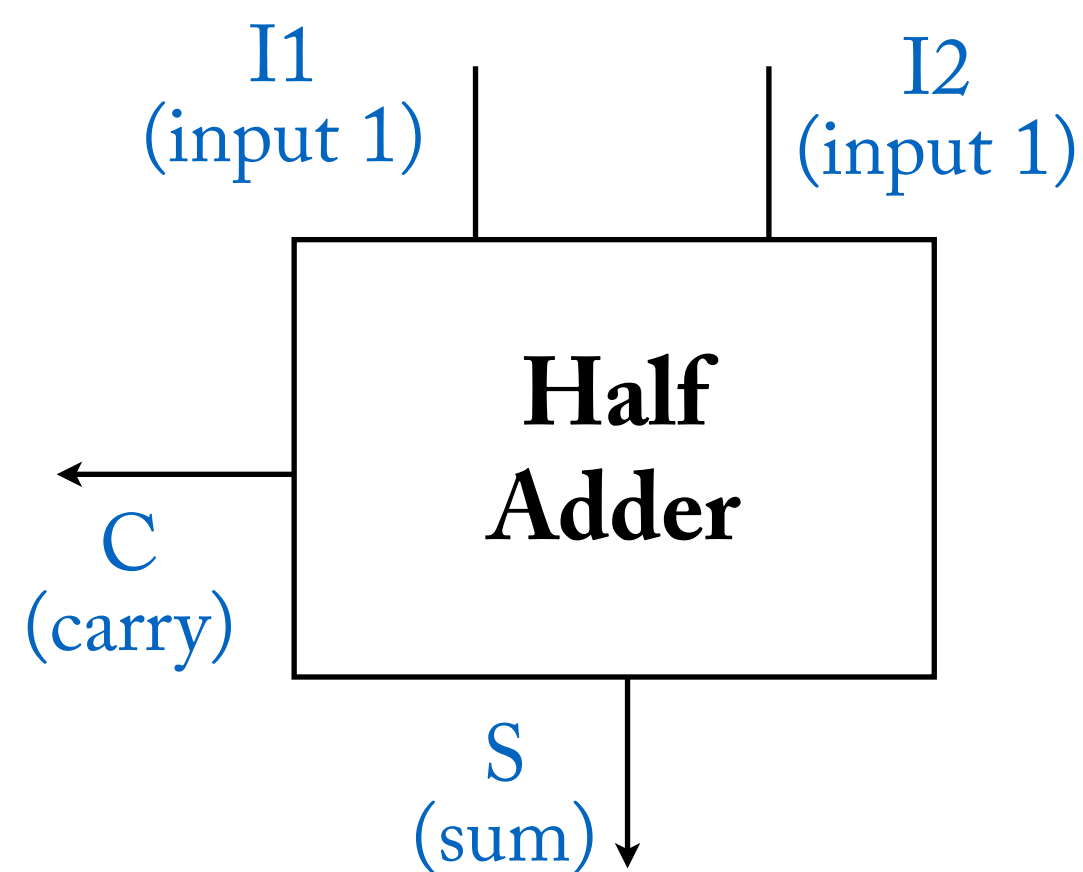


8. Functional Abstraction

Diagramming abstraction

Once you've shown how to build an XOR and Half-Adder from logic gates you can go ahead and treat these like logic gates themselves. I'll call these **boxed** circuits. That is, you can use elements like these in your circuits:



By the way: if you use a Half-Adder, it is very important to label your "carry" and "sum" outputs.

In general, once you have defined a circuit for a given function (say, binary +1[0-7]), you can use that circuit later as part of other circuits. For example:

